



ORGANIC ACID — FULL MIND MAP

Chemistry · Chapter 7.D · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

Topics in this chapter:

1. Acid Strength, Formic & Oxalic Acid
2. Lactic Acid, Muscle Fatigue & Curd Formation
3. Common Organic Acids & Natural Sources
4. Important Uses — Photography & Baking Powder
5. Acetic Acid & Vinegar
6. Optical Activity — Glycine

1. Acid Strength, Formic & Oxalic Acid

- Formic acid (HCOOH) aur Acetic acid (CH_3COOH) dono organic acids hain, lekin Formic acid stronger hota hai. Acetic acid ka $-\text{CH}_3$ group electron-donating effect deta hai, jis se H^+ release comparatively difficult hota hai; isi liye Assertion aur Reason dono true hain, par Reason strength ka correct explanation nahi hai. [Q1]
- Oxalic acid ka formula $\text{C}_2\text{H}_2\text{O}_4 = (\text{COOH})_2 = \text{HOOC}-\text{COOH}$ hota hai. Iska IUPAC naam Ethanedioic acid hai aur ye simplest dicarboxylic acid maana jata hai. [Q2]

2. Lactic Acid, Muscle Fatigue & Curd Formation

- Exam-oriented conventional explanation ke mutabik prolonged strenuous exercise ke baad muscles me Lactic acid [$\text{CH}_3\text{CH}(\text{OH})\text{COOH}$] accumulate hone se fatigue/cramp-like sensation associate ki jati hai. [Q3–Q5]
- Fresh milk me Lactose milk-sugar hota hai. Lactobacillus bacteria lactose ko ferment karke Lactic acid banata hai; isi acidification se milk sour hota hai aur milk proteins coagulate hoke curd bante hain. [Q10, Q13–Q17]
- Isliye Sour milk/Curd ka key acid = Lactic acid. “Lactose” acid nahi, sugar hai; aur Acetic acid vinegar ka acid hai. [Q10–Q17]



EXAM TRAP — Q3–Q5 — “Lactic acid causes fatigue” ko exam-language samjho

- Traditional exam texts muscle fatigue ko lactic-acid accumulation se link karte hain, isliye source-key ke liye Lactic acid yaad rakho.
- Modern exercise physiology me fatigue multi-factorial hoti hai; lactate ko single direct cause maanna oversimplification hai. Competitive-exam answer aur deeper physiology ko mix mat karna.

3. Common Organic Acids & Natural Sources

- High-yield source pairs: Lemon/Citrus → Citric acid; Vinegar/Pickle → Acetic acid; Milk/Sour milk → Lactic acid; Rancid butter/Butter → Butyric acid; Apple → Malic acid; Cold drinks/Soda water → Carbonic acid. [Q6–Q9, Q12]
- Ascorbic acid = Vitamin C aur fresh fruits/vegetables se associated hai; Formic acid red ants/bee-sting context me aata hai. Curd me Lactic acid hota hai, Acetic acid nahi. [Q11]
- Grapes ka predominant organic acid Tartaric acid hai; grapes, tamarind aur banana me bhi Tartaric acid exam-pair ke roop me poocha gaya hai. [Q18, Q20]
- Tomato me mainly Citric acid aur Malic acid milte hain; Oxalic acid comparatively kam amount me ho sakta hai. [Q19]

4. Important Uses — Photography & Baking Powder

- Source ke according Oxalic acid photography-related processing/cleaning me reducing agent ke roop me use ho sakta hai. Direct exam pair: Photography → Oxalic acid. [Q21]
- Tartaric acid baking powder me acidic component ke roop me use hota hai; Baking soda (NaHCO_3) ke saath react karke CO_2 release karne aur leavening me help karta hai. [Q22]



EXAM TRAP — Q21 — Photography ka “main chemical” mat samajhna

- Source answer Oxalic acid hai, jo kuch photographic processing/cleaning contexts me use ho sakta hai.
- Photography chemistry me silver-halide compounds bhi fundamental hote hain; isliye “photography = only oxalic acid” jaisa broad statement mat banana.

5. Acetic Acid & Vinegar

- Acetic acid (CH_3COOH), jise Ethanoic acid bhi kehte hain, vinegar ke sour taste aur pungent smell ka main reason hai. Vinegar basically dilute aqueous acetic-acid solution hota hai. [Q23–Q28]
- Vinegar food preservative aur cooking ingredient ke roop me use hota hai. Sugar-rich raw material se pehle ethanol banta hai aur phir acetic-acid bacteria ethanol ko acetic acid me oxidise karte hain. [Q24, Q26, Q29]
- Q29 me A, B aur D correct hain: vinegar fermentation se ban sakta hai, usme Acetic acid hota hai aur preservative ke roop me use hota hai; Benzoic acid vinegar ka chief acid nahi hai. [Q29]



EXAM TRAP — Q27–Q28 — “Vinegar ka chemical name = Acetic acid” ek exam-shortcut hai

- ▶ Pure Acetic acid aur Vinegar same cheez nahi hain: Vinegar ek dilute aqueous solution/food product hai jiska chief acid Acetic acid hota hai.
- ▶ Exam me option-based question ho to Acetic acid/Ethanoic acid select hota hai, lekin conceptually vinegar ko pure compound mat samjho.

6. Optical Activity — Glycine

- Optical activity ke liye molecule me generally chiral carbon chahiye jo four different groups se attached ho. Glycine ($\text{NH}_2\text{—CH}_2\text{—COOH}$) me α -carbon par do H atoms hote hain, isliye wo achiral aur optically inactive hota hai. [Q30]

✓ Source-question mapping preserved: Q1–Q30